

# **EMAIL SPAM DETECTOR: CHROME EXTENSION WITH PYTHON BACKEND**

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## **Course**

Course Name: 4495 – Applied Research Project

Section: 002

## 1. Introduction

Spam mails are among the biggest threats in cyber security as it cause loss of productivity apart from increasing the risk of users falling victims to phishing and malware among other risks. This project will ensure the archiving of an efficient real-time spam detection through the development of Chrome Extension coupled with a Python backend. In contrast to a number of tools based on the server, this utility works in the browser and is more convenient and fast. It is an extension that extracts the content of an email and transfers it to the backend to generate and display the classification immediately. This paper presents the current design and the changes made to the system and the API back end, the manifest file, and the front end of the system design for email security and proper functionality.

## 2. System Overview

The Spam Detection Chrome Extension with Real-Time Email Scanning is designed to enhance email security by automatically analyzing incoming emails in Gmail to detect spam messages. The system consists of two main components:

### Chrome Extension:

Uses **content.js** to monitor Gmail for new emails and extract subject lines and snippets.

Communicates with **background.js**, which sends email data to a backend API for analysis.

Highlights potential spam emails in red within the Gmail inbox.

### Backend API (Flask-based):

- Processes email text using a machine learning model to classify spam vs. non-spam.
- Runs locally but can be deployed for public access using ngrok or cloud platforms.
- Uses joblib to load the trained spam\_model.pkl file.

## **Technology Stack:**

- Frontend (Chrome Extension): JavaScript (content & background scripts), HTML, CSS
- Backend (Spam Detection API): Python, Flask, scikit-learn, joblib
- Machine Learning Model: Trained on spam datasets using TfidfVectorizer + Naïve Bayes
- Deployment & Testing: GitHub, ngrok (for public API testing)

The system provides a seamless real-time solution for detecting spam in Gmail while maintaining a lightweight and user-friendly interface.

## **Description of Work Done**

This week, significant progress was made in upgrading the Spam Detection Chrome Extension to support real-time email scanning in Gmail. The following key components were implemented:

### **Chrome Extension Enhancements:**

- Modified Manifest.json to request Gmail permissions.
- Updated content.js to monitor incoming emails using MutationObserver.
- Configured background.js to send extracted email data to the Flask API.

### **Backend API Development:**

- Created a Flask API to handle spam classification.
- Integrated the trained spam detection model (spam\_model.pkl).
- Encountered and resolved issues related to pickle vs. joblib compatibility.

### **Testing & Debugging:**

- Addressed Chrome extension UI bugs.
- Debugged slow API response times and identified areas for optimization.
- Encountered scikit-learn version conflicts, leading to retraining considerations.

## **Current Status:**

Some errors persist, mainly related to model compatibility and package installations. Work is ongoing to ensure seamless API response and proper email classification.

## Work Date/Hours Logs

| Date         | Member        | Hours | Descriptions                                                                                                                           |
|--------------|---------------|-------|----------------------------------------------------------------------------------------------------------------------------------------|
| Jan 14 2025  | Anchal Anchal | 2.4   | Started searching for the project topic                                                                                                |
| Jan 15 2025  | Anchal Anchal | 2     | Looking for suitable topic                                                                                                             |
| Jan 17 2025  | Anchal Anchal | 3     | Collected Information and made document about the fundamentals                                                                         |
| Jan 20 2025  | Anchal Anchal | 4     | Got some ideas from professor and started looking forward on project                                                                   |
| Jan 22 2025  | Anchal Anchal | 4     | Search for the tools and language to use                                                                                               |
| Jan 23 2025  | Anchal Anchal | 3.5   | Started working on the project, created json files for plugin                                                                          |
| Jan 25 2025  | Anchal Anchal | 3.5   | created html and css pages for the chrome extension and some changes in chrome plugin                                                  |
| Jan 27 2025  | Anchal Anchal | 4     | studied some topics from machine laeaning and tried to implement in the project                                                        |
| Jan 29 2025  | Anchal Anchal | 3     | Faced diccifulty to push files to github and imported libraries                                                                        |
| Jan 31 2025  | Anchal Anchal | 3.5   | Learned Machine laeaning Bay's theorem and did some practice to implement in the project                                               |
| Feb 02 2025  | Anchal Anchal | 4     | Learned how to create chrome extension and tried to create it                                                                          |
| Feb 03 2025  | Anchal Anchal | 3     | Created chrome extension files and uploaded to chrome extension                                                                        |
| Feb 05 2025  | Anchal Anchal | 2.5   | Found out errors encountered while data cleaning                                                                                       |
| Feb 6 2025   | Anchal Anchal | 3     | Pushed files to github                                                                                                                 |
| Feb 9 2025   | Anchal Anchal | 1     | Created Report and checked the files done                                                                                              |
| Feb 12,2025  | Anchal Anchal | 2     | Installed libraries like blinker click and colorama                                                                                    |
| Feb 16,2025  | Anchal Anchal | 2     | imported metadata worked on api                                                                                                        |
| Feb 17,2025  | Anchal Anchal | 3     | Set ted Up the Backend (Flask API)                                                                                                     |
| Feb 23,2025  | Anchal Anchal | 3     | Pushed files on github and made video to upload.                                                                                       |
| Feb 25,2025  | Anchal Anchal | 4     | Studied Chrome extension architecture and reviewed Gmail content script integration.                                                   |
| Feb 28,2025  | Anchal Anchal | 3.5   | Created Manifest.json, added permissions for Gmail, and set up basic Chrome extension structure.                                       |
| March 2,2025 | Anchal Anchal | 3     | Started development of content.js for scanning Gmail inbox.<br>Implemented email extraction from Gmail UI and tested script injection. |

|                |               |      |                                                                                                                                                     |
|----------------|---------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| March 4,2025   | Anchal Anchal | 4    | Worked on background.js to handle email content processing and API communication.                                                                   |
| March 6,2025   | Anchal Anchal | 3.75 | Developed Flask API to receive email data for spam classification.                                                                                  |
| March 7,2025   | Anchal Anchal | 4    | Saved trained model as spam_model.pkl and integrated with Flask API again.                                                                          |
| March 9,2025   | Anchal Anchal | 4.2  | Debugged API integration, identified issues with joblib vs pickle loading. Implemented MutationObserver in content.js to detect live email updates. |
| March 10 ,2025 | Anchal Anchal | 3    | Conducted initial testing, discovered API response lag issues. Optimized API response time by refining text preprocessing steps.                    |
| March 12 2025  | Anchal Anchal | 3    | Implemented email highlighting feature for detected spam messages. Deployed Flask API temporarily using ngrok for public access testing.            |
| March 13 ,2025 | Anchal Anchal | 4    | Debugged scikit-learn version conflicts, considered model retraining. Refined spam classification pipeline, improved accuracy metrics.              |
| March 15,2025  | Anchal Anchal | 3.75 | Integrated all extension components and tested Gmail real-time scanning. Conducted debugging and performance tuning for API interactions.           |
| March 16,2025  | Anchal Anchal | 5    | Pushed files to git and made report. Tried to resolve errors and run API.                                                                           |

## Repo Check-In of Implementation:

The following files/folders have been checked into the GitHub repository under the Implementation folder:

- Manifest.json – Updated permissions for Gmail access.
- content.js – Injected script to extract email data.
- background.js – Handles communication with Flask API.
- popup.html & popup.js – UI enhancements for manual scanning.

- app.py – Flask API for spam detection.
- spam\_model.pkl – Machine learning model for classification.

Additional work logs and debugging notes have been pushed under the DocumentsAndReports folder.